





Course Syllabus 2026/2027
Programming Algorithms (IF201)

Mohamad Hilmy Zhafrani

Course 1: Introduction
Institut Teknologi Tangerang Selatan
Semester Ganjil 2026/2027

Course Description

- This course develops logical and systematic thinking to design, implement, and evaluate algorithms.
- Topics include programming logic, control structures, arrays, strings, functions, recursion, stacks, and queues.
- We also discuss debugging, algorithm efficiency, and introductory applications in data analysis and AI.
- Learning activities include discussion, programming exercises, and a simple project.

IF201, 2 credits (2 theory + 0 practicum), semester 1, compulsory, no prerequisites.

Course Learning Outcomes

- CPMK-IF201-1: Explain the basic concepts of algorithms and programming.
- CPMK-IF201-2: Develop algorithms to solve computational problems.
- CPMK-IF201-3: Implement algorithms in a programming language.
- CPMK-IF201-4: Test and improve simple programs.

Related program outcomes: CPL-03 (computing concepts) and CPL-08 (implementation).

Lecturer



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Zhafran works in information systems development and serves as a civil servant in the South Tangerang City Government. His experience includes IT consulting, entrepreneurship, and education.

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Grading System

Component	Percentage
Mid-term Examination	15%
Final Examination	20%
Attendance / Quizzes	10%
Practice / Assignments (Assignments 1–6)	30%
Presentation (Assignment 7)	15%
Group Assignment (Assignment 8)	10%
Total	100%

Assignments 1–6: 5% each. Assignment 5 is a group activity.

How Will We Learn?

- Attend at least 12 of the 14 teaching sessions to be eligible for the final examination.
- Be on time. Arrival after the announced limit counts as an absence.
- Each teaching week includes 100 minutes of class, 120 minutes of structured work, and 120 minutes of independent study.
- Take part in discussions, explain your reasoning, and test your solutions.

Course Agenda (1)

Week	Topics	Assessment
1	Introduction to Algorithms and Programming	Attendance / Quiz 3%
2	Algorithm Representation	Assignment 1 (5%)
3	Basic Elements and Program Logic	Attendance / Quiz 3%
4	Structured Solution Design	Assignment 2 (5%)
5	Control Structures and Modular Design	Assignment 3 (5%)
6	Data Structures and Modular Solutions	Attendance / Quiz 2%
7	Basic Algorithm Implementation	Assignment 4 (5%)
8	Mid-term Examination	15%

Course Agenda (2)

Week	Topics	Assessment
9	Implementing Arrays and Functions	Attendance / Quiz 2%
10	Debugging and Program Tracing	Assignment 5 (5%)
11	Problem-solving Strategies and Applications	Assignment 7 (15%)
12	Algorithm Evaluation and Integrated Project	Assignment 8 (10%)
13	Introduction to Machine Learning and Data Analysis	Assignment 6 (5%)
14	Project Consultation and Further Practice	Feedback
15	Review and Final Examination Preparation	Feedback
16	Final Examination	20%





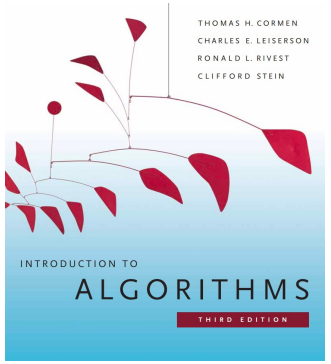


What Are Algorithms and Why Should You Care?

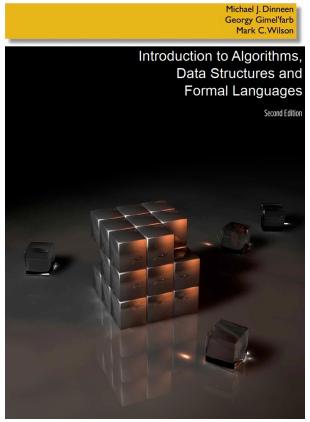
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Prescribed Texts



Cormen (2009)
Introduction to Algorithms
3rd edition



Dinneen (2008)
Introduction to Algorithms,
Data Structures and Formal Languages



Cormen (2013)
Algorithms Unlocked

Supporting references

Hidayat (2022), Algoritma dan Pemrograman, ISTN.
Kautsar (2017), Algoritma & Pemrograman – Python Series, UMSIDA Press.

Objectives

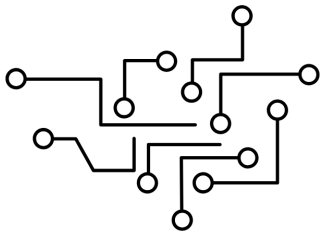
- To understand the definition and basic concept of algorithms.
- To identify the characteristics of an algorithm.
- To understand the role of algorithms in solving computational problems.
- To explain the difference between an algorithm and a program.

SUB-CPMK-IF201-1-1: Identify the main concepts of algorithms and programming.

What is an Algorithm?

- An algorithm is a clear, finite sequence of steps to solve a problem or perform a task.
- What data do we need? This is the input.
- What steps do we perform? This is the process.
- What result do we expect? This is the output.

Does an algorithm always need a computer?



A Simple Example: Shopping

- Suppose you buy 3 items at Rp5,000 each. How do you calculate the total?
- Input: the price per item and the quantity.
- Process: read the input, multiply price by quantity, and display the result.
- Output: Rp15,000.

What would change if the shop offered a discount?

Algorithms in Your Daily Life

- Can you name an algorithm you use every day without realizing it?
- What input does the activity need?
- What steps do you follow, and in what order?
- How do you know when the activity is complete?
- Could someone else follow your steps and obtain the expected result?

Characteristics of an Algorithm

- Finiteness: it ends after a finite number of steps.
- Definiteness: each step is clear and unambiguous.
- Input: it accepts zero or more inputs.
- Output: it produces at least one output.
- Effectiveness: each step can actually be carried out.



“Repeat until it looks good.” What is missing from this instruction?

Role of Algorithms in Computing

- Algorithms guide how software solves a problem.
- Searching: finding a book or a product in a list.
- Sorting: arranging prices from lowest to highest.
- Navigation: finding a route through a road network.
- Data analysis: preparing data and calculating useful summaries.



What problem does each algorithm help the user solve?

What if an Algorithm is Inefficient?

- Suppose you need to find one name in a long list.
- One approach is to check every name from the beginning.
- If the list is sorted, can you reduce the number of names to check?
- What happens as the amount of data grows?
- How would you compare the time and memory used by different solutions?

Algorithms and Programming

- An algorithm describes the logic. A program implements that logic in a programming language.

Algorithm: area of a square

- Set side to 5.
- Calculate area = side × side.
- Display the area.

Simple Python example

```
side = 5
area = side * side
print(area)
```

Output: 25

Could you implement the same algorithm in another language?

Quick Check

- How would you define an algorithm in your own words?
- Which characteristic is missing in “repeat until it looks good”?
- An algorithm calculates a rectangle’s area using length 4 and width 3. What are the input, process, and output?
- What is the difference between an algorithm and a program?

Week 1 assessment: attendance, participation, and a short quiz (3% combined).

Week 1: Structured Work

- Choose a simple problem from an activity you encounter.
- Write a short summary of its goal, input, steps, and expected output.
- Check whether the steps are clear and the algorithm stops.
- Try one set of input values. Does the result match your expectation?
- Use the same case to practise algorithm representation next week.

Structured work: 120 minutes. Independent study: 120 minutes.

Closing Remarks

- What is your view of algorithms now?
- Why do clear steps matter when solving a problem?
- Can a program run and still give the wrong result?
- Give an example of an algorithm and explain how it satisfies the key characteristics.



Next topic: Algorithm Representation

Narrative descriptions, pseudocode, and flowcharts.

Thank You

Terima Kasih

Matur Kaseh

Hatur Nuhun

Kurre Sumanga

Matur Suwun

Tarimo Kasih

Tanki Dapa

Suksma

Wa…Wa

Nggahi Weki

Mbore

Nuhan

Mauliate

Tabea

Ami Ibi

Tinggilan

Ofo Nai

Ti Mopohula To

Kando Kami

Maita Komo

Kaseh Lima